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THEME–A

India and the World, Land and the People

Our planet is a vast and fascinating place filled with diverse regions, cultures, and natural wonders. Every place on Earth has its own unique features that influence the way people live, work, and connect with one another. India, with its rich heritage and varied landscapes, is an important part of this global story. From high mountains and wide plains to coastal areas and islands, different surroundings shape human life in many ways. By exploring the Earth and its many regions, we learn how nature and people are closely connected and understand the beauty and diversity of our world.

1 Locating Places on the Earth

Let's Learn About

- A Map and Its Components
- Mapping the Earth
- Understanding Time Zones

Let's Gear Up

Imagine a big globe turning slowly inside a classroom. Children stand together, looking at bright continents. One child follows the lines of latitude and longitude, learning how these special lines help us find correct places. Another points to the equator, the warm line around Earth, while her friend finds the cold poles at the top and bottom. The teacher opens an atlas, showing maps with countries, cities and famous places. The Eiffel Tower in Paris, the Pyramids in Egypt and the Great Wall of China seem exciting on the pages. A compass is shared around, teaching directions-north, south, east and west. Together, they learn that finding places is like solving a puzzle: maps, globes and co-ordinates are the clues. With these tools, everyone can explore the world through imagination, finding mountains, rivers and oceans, while understanding how our planet stays connected.

In this story, we learn how maps and directions help us find different places. Now let's learn about maps and directions more clearly together.

A MAP AND ITS COMPONENTS

The word "Map" comes from the Latin word "mappa," which means "sheet" or "napkin." It is called this because the first maps were usually drawn or made on cloth, skin and parchment. If we draw the world or any part of it on paper, that picture is called a map.

Thus, a map is a drawing or picture of the Earth's surface or any part of it on a flat surface using a scale.

Types of Maps

Broadly, maps can be divided into three types:

(I) Physical or Relief Maps: Maps that show Earth's natural features like mountains, plateaus, plains, rivers and oceans are called physical or relief maps.

Physical Map of India

Political Map of India

(ii) Political Maps: Maps that show cities, towns, villages, and different states and countries with their boundaries are called political maps. For example, a map of India showing all the states is a political map.

(iii) Thematic Maps: Maps based on a particular theme are called thematic maps. They show special information, such as climate, rainfall, railways, roads, farming, and more.

In-Text Questions

1. "Mappa" is a ..... word.
2. Rivers and oceans are shown in ..... maps.
3. Maps showing climate, rainfall or population are called ..... maps.
4. A map that shows states and borders is a ..... map.

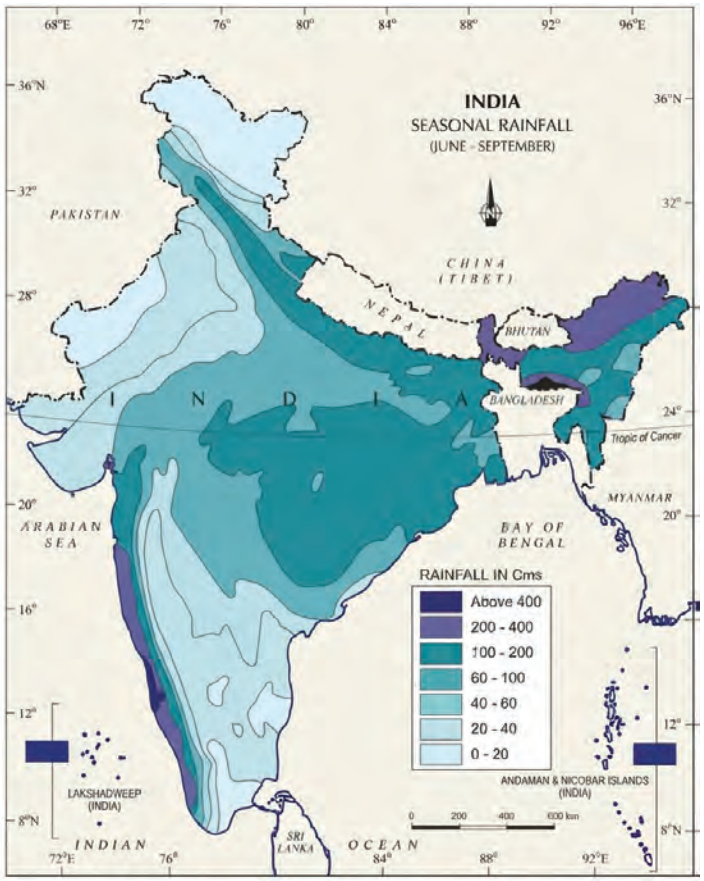
Do you know?

- A book that contains maps of the world, countries, and continents is called an Atlas.
- On 20 May 1570, Abraham Ortelius, a Flemish mapmaker, published "Theatrum Orbis Terrarum," the world's first Atlas.

Abraham Ortelius

Atlas

- A cartographer is a person who makes or draws maps.
- Cartography is the art and science of creating and drawing maps.



Thematic Map (India Seasonal Rainfall)

Components of a Map

A map has four main parts. Without these parts, we cannot understand a map properly. They give us useful information about the map.

The four components are:

Title

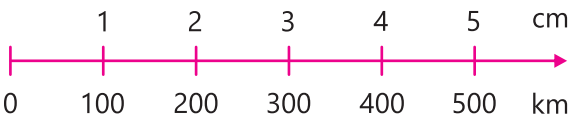
The title gives the name of a map. It helps the reader understand what the map shows. Every map should have a clear and proper title.

For example: if a map is named "Population of India," it shows how people are living in different areas of India.

Scale

Scale means the relation between a distance on a map and the real distance on land. Distances on maps are found by using the scale. Every map is made with a scale. A scale helps maps become smaller or larger. The size of a map is shown in comparison with the size of a place through a scale. *For example:* When the distance between place A and place B is 100 km in reality, it is shown as 1 cm on the map, which means 100 km = 1 cm on the map. This shows that 100 km on the ground equals 1 cm on the map.

Depending on the size of the scale, maps are divided into different types:



Scale : 1 cm= 100 km.

(i) Small-Scale Maps: Maps that use a small-scale to show big areas of the Earth are called small-scale maps, for example, 1 cm = 100 km. They give fewer details about large regions like continents or countries. The maps in your atlas are made using a small-scale. The map of India shown in your book is also an example of a small-scale map.

Scale : 1 cm= 100 m.

(ii) Large-Scale Maps: Maps made with a large scales are called large-scale maps. These maps show small areas on a bigger-sized map. Their main purpose is to show many details about a small area, for example, 1 cm = 100 metres.

A map of a village or city is an example of a large-scale map.

The differences between small-scale maps and large-scale maps can be understood in these ways:

Small-Scale Maps	Large-Scale Maps
1. These maps show big parts of the Earth's surface (1 cm = 100 km).	1. These maps display small parts of the Earth's surface (1 cm = 100 m).
2. They display only limited details about a large area.	2. They show a large amount of information about a small area.
3. A world map is a small-scale map. All the maps shown in an atlas are small-scale maps.	3. A road map, city map etc. are examples of large-scale maps.

Direction

Have you ever noticed an arrow with the letter 'N' at the corner of a map? It shows the north direction. Most maps use an arrow marked with 'N' to show directions. Some maps also have a compass to help us find directions.

Compass

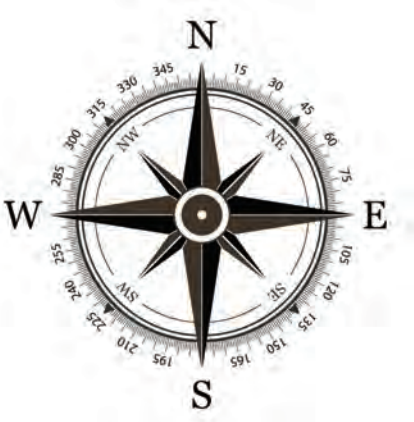
Direction is the way in which a person or object moves. A compass is a tool with a magnetic needle that helps us find directions. Usually, directions are divided into two major types.

INDIA - Outline Map

Map showing North Direction

Cardinal Directions

North (N), South (S), East (E) and West (W) are the main directions. They are also called the four major points of a compass.



Intermediate Directions

Intermediate Directions

Directions found between two main points are called middle directions. North-East, South-East, North-West and South-West are these directions.

All these directions help us find the right way. They help us locate places easily.

Cardinal Directions

Conventional Symbols

Conventional symbols are common signs used on maps to show different features of the Earth's surface. These symbols help us know mountains, rivers, roads and buildings in small spaces. They make the map simple to read and understand clearly.

Conventional symbols are called map legends because they tell us what the map shows. Like a legend, they are also called map keys because they explain various symbols.

Analyze Thoughts

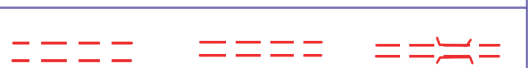
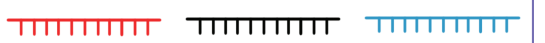
- On your way back home from school, how many turns do you make ?
- Which directions do these turns face on your way?

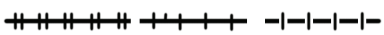
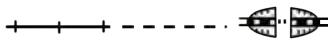
Analyze Thoughts

- Pick any four features from the table below that are found around you.

Conventional symbols are grouped into three main types:

1. Line Symbols: roads, railways, borders etc.
2. Point Symbols: wells, buildings etc.
3. Area Symbols: villages, farms etc.

Features	Symbols
Roads metalled	
Roads unmetalled	
Cart track, pack track and pass; footpath with a small bridge.	
Streams: with path in the bed; canal.	
Dams: masonry or rock-filled; earthwork, weir	
River bank: dry with water channel; with island & rocks, tidal river	
Submerged rocks, shoal; swamp, reeds	
Wells: lined, unlined; tubewell, spring; tanks: permanent, seasonal.	
Embankments: road or railway; tank; uneven ground.	
Railways: broad gauge; double; single with station; being constructed.	

Features	Symbols
other gauges; double; single with milestone; under construction.	
Mineral or tramway line, telegraph line; hill cutting with tunnel.	
Contours with smaller features; rocky slopes, steep cliffs.	
Sand features: (1) flat, (2) sand hills and dunes (mapped), (3) moving dunes.	
Towns or villages: populated; abandoned; fort.	
Huts: permanent; temporary; tower; historical remains.	
Lighthouse, lightship, buoys–lighted or unlighted, anchorage.	
Temple, chhatri, church, mosque, idgah, tomb, cemetery.	
Mine, grass and scrub	
Palms: palmyra; other plantations; conifers; bamboos; other trees.	
Surveyed tree	
Boundary: international	
Boundary: state: demarcated; undemarcated	
Boundary: district; subdivision; Tehsil or Taluk; forest	
Boundary pillars: surveyed; unlocated; village trijunction	
Heights: triangulated station; point; approximate	△ 200 • 200 • 200
Beench-mark: geodetic; tertiary; canal	• BM 63.2 • *BM 63.2* • 63
Post office; telegraph office, combined office, police station	*PO* *TO* *PTO* *PS*

In-Text Questions

1. What are the three main types of maps?
2. Draw a school map showing the staffroom, the principal's office, the library and the playground, using directions, a scale and colours to label areas clearly and properly.
3. Find out :
   (a) How many centimetres are there in one kilometre?
   (b) How many inches are there in a single mile?
   (c) How many metres are there in one kilometre?

Colour codes help show various types of land features.

Colour	Features
Brown	Land features like mountains
Black	Roads, railways and boundaries
Blue	Water
Red	Settlements and construction features
Green	Vegetation/plains
Yellow	Plateaus

Go Further

Compare the differences between a globe and a map :

	Globe	Map
1	A globe is shaped like a round sphere.	A map is a flat representation of the Earth's surface.
2	A globe is three-dimensional in shape	A map is two-dimensional.
3	A globe shows the correct shape of continents and the distances between lands.	A map distorts the shape and size of continents.

MAPPING THE EARTH

Maps are useful when we need to study a part of the Earth, such as a country, state or city. But how can we study the whole Earth? We know that the Earth is round, while maps are flat. A flat map cannot show the Earth's curved surface properly without making some changes. To study the entire Earth, people made a small and correct model to view it fully. This model of the Earth is called a globe.

A globe is a correct model or a small copy of the Earth. It is a hollow ball that shows the whole world, including all lands and oceans. Its shape is similar to the Earth's, so it shows the actual size and shape of continents and oceans. It also shows different landforms and surface features.

It shows features like hills, plateaus and plains. It helps to find the shortest distance between places and shows the exact location and direction of places compared to each other.

If you observe the globe carefully, you will see many horizontal and vertical lines.

A Globe

Now, let us learn about some parts of the globe:

(a) Understanding Coordinates

We have learned that a globe is better for finding any place on Earth because its shape is similar to the Earth's shape. To locate any place on Earth correctly, we use the coordinate system. A geographic coordinate system is a three-dimensional method used to find places on the Earth's surface. The exact address of any place on Earth is shown by two numbers: one for latitude and one for longitude. These two numbers are called coordinates. Longitude shows the east-west position, while latitude shows the north-south position of a place. In this way, we can find the exact location of any place on Earth.

It is very hard to find and locate these features without using any fixed point.

Learn about a reference point by seeing the following example:

Imagine that today is your birthday and you have invited your friends. They do not know the location of your house. You tell them that your house is near D.A.V. School, opposite Prabhat Bakery. Here, you use D.A.V. School and Prabhat Bakery as reference points to explain your home's location. Both of these are called reference points. However, these references are useful only in nearby areas. For national and international purposes, we need references that are accepted worldwide. Latitudes and longitudes are examples of such reference points. Now, let us learn about latitudes and longitudes in more detail.

In-Text Questions

1. What do we mean by a globe?
2. Pick up a globe and locate your country, India, on it.
3. Write a list of objects that are shaped like the Earth.

(b) Latitudes

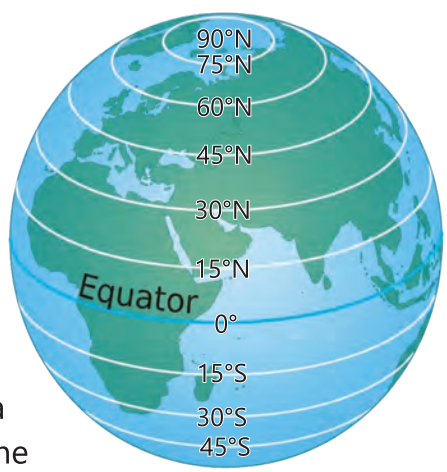
Latitude is the distance of a place on Earth measured in degrees from an imaginary line called the equator, which is located midway between the North and South Poles.

Usually, distance is measured in kilometres or miles, but angular distance is measured in degrees. On the globe, you can see lines going from east to west. These lines are imaginary and are called latitudes.

Key Facts about Latitudes

- Latitudes are parallel to each other. All circles that are parallel to the equator up to the poles are called parallels of latitude.
- These are imaginary lines going from east to west, used to measure north-south positions on Earth.
- The gap between two latitude lines is considered 1°, which is equal to 111 km.

  The Equator is the most important latitude. It is used as a reference because it divides the globe into two equal parts. The upper half of the Earth is called the Northern Hemisphere and the



Latitudes

lower half is called the Southern Hemisphere. The Equator shows zero degrees latitude and is the biggest latitude on Earth.

Northern Hemisphere
Southern Hemisphere
↑ → 0° Latitude (Equator)
↓

- All lines above the Equator are called north latitudes. Similarly, all lines below the Equator are called south latitudes.
- The distance from the Equator to the North or South Pole is one-fourth of a circle around the Earth, which is 1/4 of 360°, or 90°. Therefore, 90°N represents the North Pole and 90°S represents the South Pole.
- The letter 'N' shows North latitudes and the letter 'S' shows South latitudes on maps and globes.
- Not all latitude lines have the same length. Their length decreases as you move closer to the poles.
- The gap between any two latitude lines stays the same everywhere.

Go Further

The circumference of the Earth is 40,000 km and being spherical, it has a full 360° angle.

For 360° the distance is 40,000 km (Earth's circumference).

For 1° = 40,000/360 = 111.11 km

Thus, 1° of latitude equals about 111 km on Earth.

Some Important Latitudes

(i) The Equator: Represents 0° latitude.

(ii) The Tropic of Cancer: Located at 23½° north in the Northern Hemisphere.

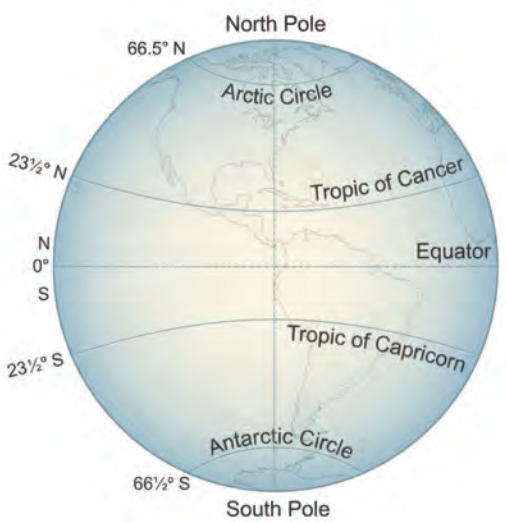
(iii) The Tropic of Capricorn: Located at 23½° south latitude in the Southern Hemisphere.

(iv) The Arctic Circle: Located at 66½° North latitude in the Northern Hemisphere.

(v) The Antarctic Circle: Located at 66½° South latitude in the North Southern Hemisphere.

(vi) The North Pole: Located at 90°N in the Northern Hemisphere.

(vii) The South Pole : Located at 90°S latitude in the Southern Hemisphere.



Important Latitudes

Do you know?

In India, the Tropic of Cancer crosses eight states: Rajasthan, Gujarat, Madhya Pradesh, Chhattisgarh, Jharkhand, West Bengal, Tripura and Mizoram.

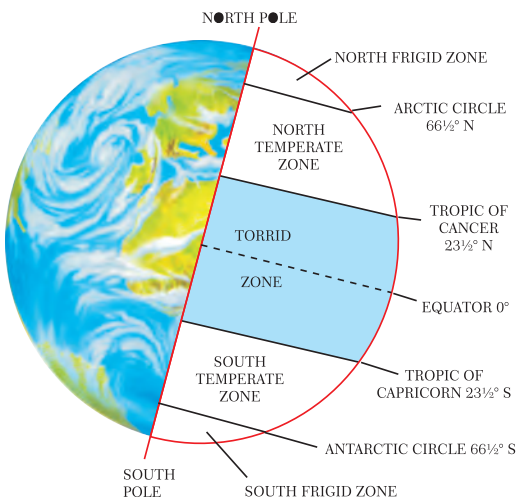
Latitudes and Climate

Different areas of the Earth receive different amounts of heat from the Sun, causing climate changes with latitude. The Earth is divided into three heat zones based on the Sun's heat.

These zones are:

(i) Torrid Zone

- Torrid means extremely hot. It is the hottest zone on the Earth.
- It is found between the Tropic of Cancer and the Tropic of Capricorn spanning across the Equator.
- The noon sun is directly overhead at least once every year at all latitudes within this zone, so it receives the most heat.



Heat Zones

(ii) Temperate Zone

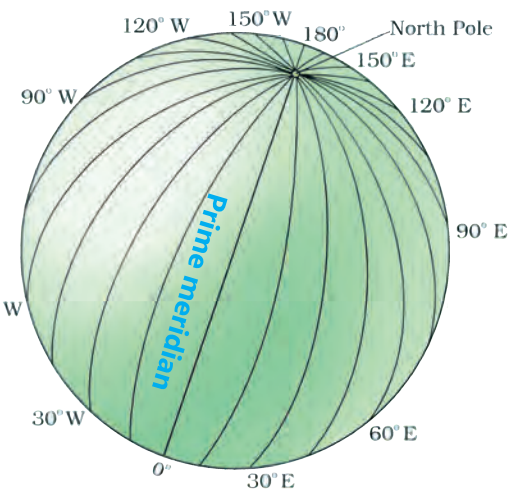
- It is found between the Tropic of Cancer and the Arctic Circle in the Northern Hemisphere and the Tropic of Capricorn and the Antarctic Circle in the Southern Hemisphere.
- These areas have moderate temperatures because they get slanting sunlight all year. This happens because the Sun's rays fall at a lower angle near the poles.

(iii) Frigid Zone

- This zone is found between the Arctic Circle and North Pole in the Northern Hemisphere and the Antarctic Circle and South Pole in the Southern Hemisphere.
- These are the coldest regions on Earth because they receive highly slanting rays from the Sun.

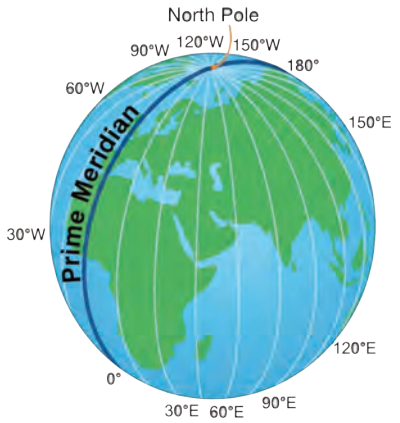
(c) Longitudes

Look at the globe carefully again. You will see many lines running from the North Pole to the South Pole. These lines are used to determine east-west positions on Earth. These lines are also called meridians of longitude.



Longitudes with Prime Meridian

Unlike latitudes, all meridians have the same length, so any meridian can be used for measuring longitude. In 1884, an international agreement chose the meridian passing through the British Royal Observatory at Greenwich, near London, as the starting line for counting longitudes. This meridian is called the Prime Meridian. Its longitude is 0°. From here, we measure 180° east and 180° west.



Longitudes

Some Important Facts about Longitudes

- The distance between two meridians is measured in degrees, which are divided into minutes and then into seconds.
- There are 360 meridians of longitude, divided equally into 180° east and 180° west.
- Longitudes are half-circle lines. They are all equal in length but are not parallel.
- The distance between two meridians of longitude is not the same. It becomes smaller toward the poles and becomes zero where all meridians meet.
- The Prime Meridian and the 180° meridian together divides the Earth into two halves: the Eastern Hemisphere and the Western Hemisphere.
- The letter 'E' shows Eastern longitudes and 'W' shows Western longitudes on maps and globes.
- The 180° East and 180° West meridians of longitude are on the same line.

Grid

As explained earlier, latitudes and longitudes cross each other on the globe to form a grid system. This network of lines serves as the Earth's complete coordinate system, allowing us to find the exact location of any place on a globe or map.

Map of India with Latitudes and Longitudes

UNDERSTANDING TIME ZONES

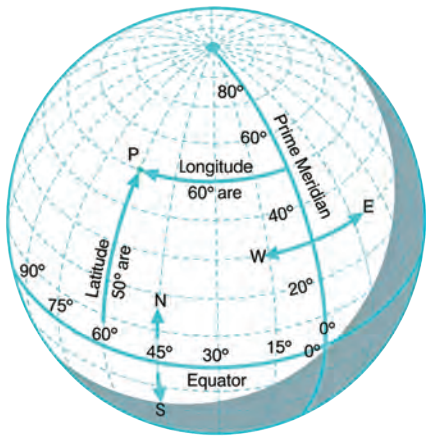
The Sun is the most reliable natural timekeeper in the world. So, time is measured based on the relationship between the Sun and the Earth's movement. As the Earth completes one rotation every 24 hours, it turns 15° per hour which is equal to 1° every 4 minutes. The Earth rotates from west to east. For every 15° eastward, local time moves ahead by 1 hour, and for every 15° westward, local time moves back by 1 hour. Therefore, places east of the Greenwich Meridian see the Sun earlier and gain

time, while places west see it later and lose time.

Local time at westward New York	Local time at GMT London	Local time at eastward India
is 7:04 am (4 hrs. 56 min. behind)	(12 noon)	is 5:30 pm (5 hrs. 30 min. ahead)
Lose time going Westwards	0° (Greenwich Meridian)	Gain time going Eastwards

Local Time

All places located on the same longitude have the same local time, which is called local time. This is calculated using the Sun's shadow, which is shortest at noon and longest during sunrise and sunset every day. For example, when the Sun is at its highest point over the Prime Meridian, all places on this meridian have noon, meaning every place along a particular longitude shares the same local time.



Longitude and Time

As the Earth keeps rotating, places at different longitudes have different times. The Earth moves from west to east. Therefore, people in the east see the Sun earlier. During its rotation, the Earth takes 4 minutes to cover 1° of longitude.

The Earth completes one full rotation of 360° in 24 hours.

In 24 hours, the Earth turns 360°. In 1 hour, it turns 360° ÷ 24 = 15°.

The Earth covers 15° in 1 hour. Time taken to cover 1° = 60 ÷ 15 = 4 minutes.

Thus, the Earth moves 1° in 4 minutes. Since the Earth takes 4 minutes to cross one meridian, it needs 24 hours, or 1440 minutes, to complete one full rotation.

Indian Standard Time (IST)

India lies between 68°7′E and 97°25′E longitudes. There are 29 longitudes between them, and we know that time changes with every longitude. If we calculate time from the Prime Meridian, India would have 29 different local times. Having many local times in India would create confusion and problems. To avoid this, the local time of a central longitude was chosen as the standard time for

the whole country.

Do you know?

What is the meaning of A.M. and P.M.?

A.M. stands for ante meridiem, meaning the time before noon.

Ante means before and meridiem means Sun's position at noon.

Thus, the time between midnight and noon is marked as A.M. For example, 6 o'clock in the morning is written as 6 A.M.

P.M. stands for post meridiem, meaning the time after noon or midday.

Thus, the time between noon and midnight is marked as P.M.

For example, 5 o'clock in the evening is written as 5 P.M.

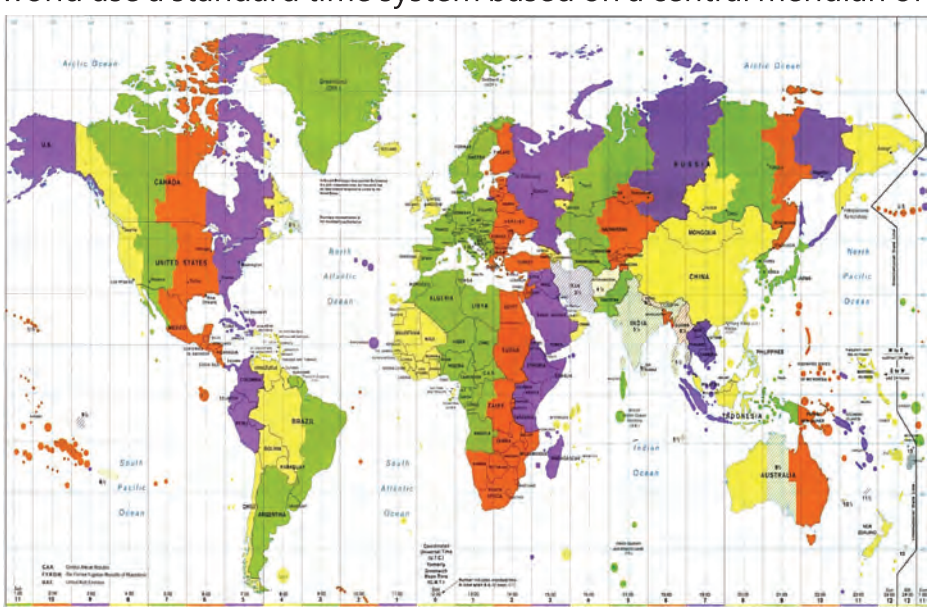
Go Further

When it is 12:00 noon in London, it is 5:30 P.M. in India because Indian Standard Time follows the local time of 82°30'E longitude. This longitude lies east of Greenwich, near London and is 5 hours 30 minutes ahead of Greenwich Mean Time (GMT).

The 82°30'E longitude, passing through Mirzapur near Prayagraj (formerly Allahabad), is chosen as the Standard Meridian. The local time along longitude is used as the standard time for the whole country. This is called Indian Standard Time, shortened as IST.

Time Zones Around the World

Like India, most countries in the world use a standard time system based on a central meridian of their area. To manage time, the world is divided into 24 standard time zones, each separated by 15° of longitude or one hour of time. Time zones are regions on Earth that have a uniform local time. They help us know the time in different places around the world. Each time zone represents one hour of the Earth's complete twenty-four-hour day. The Prime Meridian is the reference line for all time zones. The time at the Prime Meridian is called Greenwich Mean Time (GMT).



Time Zones of the World

As we move east from the Prime Meridian, the time is ahead. For every 15° of longitude we move east, one hour is added to the time. The Earth completes a 360° rotation in 24 hours, so 15° represents one hour. For example, when it is 12:00 noon at the Prime Meridian, it will be 1:00 PM at 15°E, 2:00 PM at 30°E, and so on.

But as we move west from the Prime Meridian, the time goes back or decreases. For every 15° of longitude we travel west, one hour is subtracted from the time. For example, if it is 12:00 noon at the Prime Meridian, it will be 11:00 AM at 15°W, 10:00 AM at 30°W, and so on.

Time zones help us find the time of any place on Earth. They are useful for travelling, business and communicating with others.

If you look at a world map, you will see that the lines showing time zones are not completely straight. This is because time zone lines often follow country or area boundaries. Each country or region chooses its time zone according to its needs. Time zone lines may also be adjusted to match political or administrative boundaries or to keep important cities in the same time zone for convenience.

However, large countries like the U.S.A., Canada and Russia, which extend widely from east to west, need to use many time zones. Russia covers nearly 165° of longitude and is divided into 11 time zones. This means Russia has 11 standard times. Canada and the U.S.A. each have six time zones.

The line exactly opposite the Prime Meridian, near 180° longitude, is called the International Date Line (IDL). The +12 and -12 time zones meet at this line.

When we cross the International Date Line by ship or plane, the date changes. If we travel from west to east, we move back one day. For example, the date changes from Sunday to Saturday.

But, if we travel from east to west, we move ahead one day, that is, from Sunday to Monday. The International Date Line also bends at some places to avoid dividing countries into two different dates.

By now, you should know how people in different parts of the world experience the same time differently based on their position from the Prime Meridian.

In-Text Questions

1. (a) 0° latitude represents the ..... .
   (b) The earth takes ..... to complete one rotation.
   (c) The hottest region of the earth is the ..... zone.
2. Use a map of India and mark 82½° E longitude on it.
3. Find the local time in London when it is 5:30 PM in India.
4. Find the latitude and longitude of Delhi, Mumbai and the United States.

NEW TERMS

- **Scale:** Shows real distance from map
- **Direction:** Way or path to move
- **Globe:** Round model of the Earth
- **Indian Standard Meridian:** Longitude fixing India's standard time
- **Equator:** Imaginary line dividing Earth equally
- **Map:** Flat drawing of Earth's surface
- **Indian Standard Meridian (Mirzapur):** Time reference line passing Mirzapur
- **Intermediate directions:** Directions between main directions
- **Grid:** Network of latitude and longitude lines
- **Longitudes:** Lines running from pole to pole
- **Latitudes:** Horizontal lines running east-west that measure.

QUICK TAKE

- Map essentials include title, scale, direction and common symbols.
- The Equator is an imaginary central circle that divides Earth into northern and southern halves at zero degrees.
- A map shows the Earth on a flat surface using a reduced scale to understand distances.
- Latitude measures the distance of a place north or south of the Equator in degrees.
- Types of maps include physical maps, political maps and maps showing specific information.
- Longitudes are imaginary vertical lines extending from the North Pole to the South Pole.
- The Prime Meridian passes through Greenwich Observatory in England, showing zero longitude and dividing Earth into eastern and western halves.
- India's standard time is based on 82°30′ east longitude, passing through Mirzapur near Prayagraj.
- Moving east increases time, while travelling west decreases time.
- Places on the same longitude have the same clock time, called local time.
- Earth has 360 longitude lines, equally divided into eastern and western parts.
- The globe is divided into 24 time zones, each covering 15 longitude degrees or one hour.
- The International Date Line is located opposite the Prime Meridian near 180° longitude.
- Latitude and longitude lines help find exact locations on globes or maps.

I. OBJECTIVE TYPE QUESTIONS

A. Choose the correct answer.

1. The network of lines of latitudes and longitudes is also known as a

   (a) Grid ☐ (b) Sketch ☐ (c) Plan ☐ (d) None of these ☐

2. Which map shows countries, states, districts, and other political boundaries?

   (a) Physical Map ☐ (b) Political Map ☐

   (c) Thematic Map ☐ (d) Sketch ☐

3. Directions that fall between two main cardinal points are called ..... directions.

   (a) Cardinal Directions ☐ (b) Ordinal Directions ☐

   (c) Intermediate Directions ☐ (d) Main Directions ☐

4. The distance between Place A and Place B is 1000 km on land, but it is shown as 1 cm on the map. This means 7500 km on the ground will be shown as ... cm on the map.

   (a) 75 ☐ (b) 750 ☐ (c) 0.75 ☐ (d) 7.5 ☐

5. The Antarctic Circle is located in the

   (a) Northern Hemisphere ☐ (b) Southern Hemisphere ☐

   (c) Eastern Hemisphere ☐ (d) Western Hemisphere ☐

6. Which of the following is the exact model of the Earth?

   (a) Map ☐ (b) Globe ☐ (c) Apple ☐ (d) Ball ☐

7. Which of the following statements about longitudes is false?

(a) These are semi-circular lines.

(b) The Greenwich Meridian is the most important longitude.

(c) There are 180° of longitudes.

(d) All longitudes meet at the North and South Poles.

8. Russia has how many different time zones?

(a) 5 (b) 7 (c) 8 (d) 11

9. How much time does the Earth need to move across one degree of longitude?

(a) 4 minutes (b) 6 minutes (c) 7 minutes (d) 10 minutes

10. Which of the following is called the Indian Standard Meridian?

(a) 23½°N (b) 66½°S (c) 66½°E (d) 82½°E

B. Write 'T' for True or 'F' for False.

1. A globe is a round, three-dimensional model of the Earth.
2. 66½°N latitude is known as the Arctic Circle
3. The Greenwich Meridian shows 0° longitude.
4. A map is not an exact model of the Earth.
5. Yellow colour is used to show settlements on a map.

C. Choose the option that correctly matches items from Column A and B.

Column A	Column B
1. Antarctic Circle	(i) 0° latitude
2. South Pole	(ii) 23 1/2° N
3. Equator	(iii) 66 1/2° S
4. Arctic Circle	(iv) 90°S
5. Tropic of Cancer	(v) 66 1/2°N

(a) 1.-(iii), 2.-(i), 3.-(iv), 4.-(v), 5.-(ii)

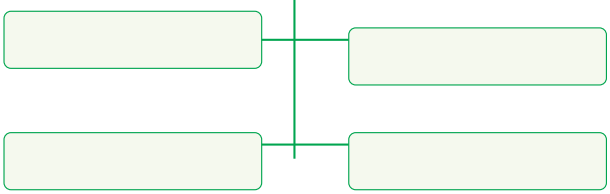
(b) 1.-(iii), 2.-(iv), 3.-(i), 4.-(v), 5.-(ii)

(c) 1.-(iv), 2.-(v), 3.-(i), 4.-(iii), 5.-(ii)

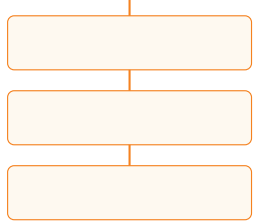
(d) 1.-(iii), 2.-(iv), 3.-(i), 4.-(ii), 5.-(v)

D. Fill in the blanks with correct answers.

1. Names of the four Components of a Map



2. Names of the three Heat Zones of Earth



E. Assertion-Reason

For question numbers 1 to 4, two statements are given—one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the codes:

1. Assertion (A) : North, South, East and West are known as cardinal directions.
   Reason (R) : Cardinal directions help us identify the main directions on a map or globe.
2. Assertion (A) : Local time varies from place to place.
   Reason (R) : Local time depends on the longitude of the place.
3. Assertion (A) : The equator divides the Earth into the Northern and Southern Hemispheres.
   Reason (R) : The equator is an imaginary line located at 0° latitude.
4. Assertion (A) : Indian Standard Time (IST) is calculated using the 82½°E longitude.
   Reason (R) : Time is the same across a time zone and does not vary with longitude.

(a) Both Assertion (A) and Reason (R) are true, and Reason explains Assertion correctly.
(b) Both Assertion (A) and Reason (R) are true, but Reason does not explain Assertion.
(c) Assertion (A) is true, but Reason (R) is false.
(d) Assertion (A) is false, but Reason (R) is true.

II. PICTURE STUDY

1. Point out the Equator on the diagram. What is its degree?
2. Show the Tropic of Cancer. Which hemisphere does it lie in?
3. Locate the Tropic of Capricorn. Which hemisphere does it lie in?
4. Find the Arctic Circle. What is its latitude value?
5. Find the Antarctic Circle. What is its latitude value?

III. SHORT ANSWER TYPE QUESTIONS

1. Name the three main types of maps.
2. What is Indian Standard Time (IST)? Why do we use one time zone for the whole of India?
3. Write any two important points about latitudes.
4. What is the Greenwich Meridian? Why is it important?

IV. LONG ANSWER TYPE QUESTIONS

1. Why are conventional symbols used on maps? How do they help us understand a map better?
2. Why is the time in India about 5 hours ahead of London? Explain in simple words.
3. What are the Torrid Zone and the Frigid Zone? Write the differences between them.
4. Explain the three main types of maps with examples.

V. SOURCE BASED QUESTIONS

Read the given paragraph and answer the following questions.

The Earth is round like a globe and to find the exact location of any place, we use a co-ordinate system. This system is based on two numbers: latitude and longitude. Latitude measures the north-south position, while longitude measures the east-west position. Together, they give the precise global address of a place. Latitudes are imaginary lines running east to west. The most important latitude is the Equator, which divides the Earth into the Northern Hemisphere and the Southern

Hemisphere. All latitudes are parallel to each other and the distance between two parallels is about 111 km. Without such references, it would be very difficult to locate places on Earth.

1. If the Equator divides Earth into two halves, what would happen if there were no Equator?
   (a) Earth would stop rotating ☐ (b) Longitude lines disappear ☐
   (c) Latitude distances would change ☐ (d) No hemispheres possible ☐
2. Why is the distance between two latitudes always about 111 km?
   (a) Earth is perfectly flat ☐ (b) Earth is spherical in shape ☐
   (c) Equator controls distances ☐ (d) Longitudes are curved ☐
3. If you only knew the longitude of a place but not the latitude, what problem would you face?
   (a) Distance cannot be measured ☐ (b) Equator cannot be located ☐
   (c) East west unclear ☐ (d) North south unclear ☐
4. Why are latitudes called 'parallels'?
   (a) Run side by side ☐ (b) Meet at poles ☐
   (c) Shorter than longitudes ☐ (d) Controlled by Equator ☐

VI. PROJECT WORK

- Students calculate time differences between India and 5 other countries.
- Present findings in a creative chart or poster.
- Discuss why time zones exist and how they affect travel/communication.

VII. GLOBE READING

Identify the following with the help of a globe.

1. Find the latitudinal and longitudinal extent of your state.
2. Identify the place in India that is surrounded by three seas.
3. Name the states through which the Tropic of Cancer passes.
4. Name the countries that make up the Indian subcontinent.
5. Locate and name the Torrid Zone.

2 Oceans and Continents

Let's Learn About

- Distribution of Water and Land on Earth
- Continents and Islands
- Oceans and Their Features

Let's Gear Up

Riya loved geography and often dreamed of becoming a great explorer. She imagined herself traveling across the huge Pacific Ocean, the biggest in the world, where endless waves shone under the sun. Her ship finally reached the incredible continent of Australia and she was happy to see kangaroos hopping freely across the grasslands. Excited to continue her adventure, Riya crossed the warm waters of the Indian Ocean and arrived in Africa, where the large Sahara Desert stretched endlessly before her eyes. She admired its golden dunes and the strength of life in such a difficult environment. Her journey did not stop there. She sailed again, this time across the Atlantic Ocean, reaching the lively lands of North America. The tall Rocky Mountains amazed her with their snowy peaks and breathtaking. Riya realized that exploring oceans and continents was like opening a treasure chest of wonders, each place offering new lessons and unforgettable experiences for everyone.

Like Riya, let's explore the wondrous world of oceans and continents!

DISTRIBUTION OF WATER AND LAND ON EARTH

Water is very important for life. Earth's surface is covered by approximately 71% water and 29% land. Since Earth has lots of water, it is called the Blue Planet or Water Planet. From

space, Earth looks blue because of oceans, seas, lakes and rivers. All these water areas together form the hydrosphere. Water is a renewable natural resource.

Most of Earth's water, 97.3%, is found in oceans and is salty. Only 2.68% is present in glaciers, ice sheets, and water below the ground. The freshwater available for human use is just 0.03%. This small supply makes water saving very important.

Intext Questions

Water Distribution	(% of total water)
Oceans	97.3
Ice caps and glaciers	2.0
Groundwater	0.68
Freshwater lakes and inland	0.009
Salt lakes	0.009
Atmosphere	0.009
Rivers	0.0001

OCEANS

Oceans are very large water areas that make up most of the hydrosphere. Their average depth is around 4 km. Ocean water keeps moving through currents, waves and tides. All oceans join together, so the surface level stays same, known as sea level, which is used as a base for measuring elevation.

Oceans

The Five Major Oceans:

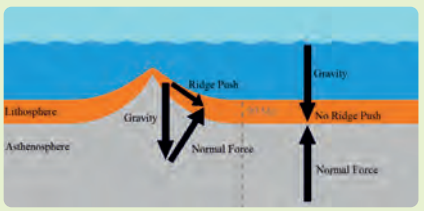
1. Pacific Ocean
2. Atlantic Ocean
3. Indian Ocean
4. Arctic Ocean
5. Southern (Antarctic) Ocean

Key Facts:

- Pacific Ocean: Largest and deepest; home to Mariana Trench (11,022 m).
- Atlantic Ocean: 'S'-shaped, long coastlines ideal for ports and trade.
- Indian Ocean: Triangular, named after India.
- Southern Ocean: Connects with Indian, Atlantic and Pacific Oceans.
- Arctic Ocean: Smallest, surrounding the North Pole, linked to Pacific via Bering Strait.

Do you know?

Underwater Mountains: More mountains exist under oceans than on land. The Mid-Ocean Ridge stretches 65,000 km, mostly underwater.



Mid-ocean ridge

World : Oceans

Importance of Oceans:

1. Provides moisture and water vapor daily.
2. Ocean currents move the sun's heat around Earth.
3. Absorb carbon dioxide, reducing global warming.
4. Separate and connect continents, aiding transport.
5. Facilitate trade and shipping routes.
6. Shelter diverse marine life.
7. Generate hydroelectricity.

Marine life

Oceans and Life:

Marine life includes plankton, fish, turtles, whales, sharks, and seabirds. Ocean plants include algae, seagrass, and seaweeds. Sharks, for example, have lived over 400 million years and can sense electromagnetic fields. Scientists estimate there are millions of marine species, with around 242,000 identified globally today.

In-Text Questions

Fill in the blanks.

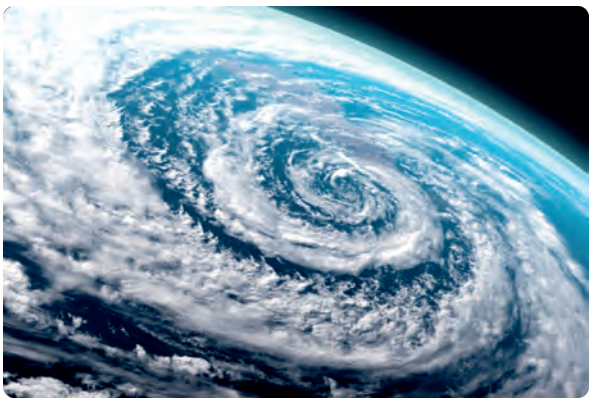
1. The average depth of the oceans is about _____ km.
2. The common level of ocean water is called _____ level.
3. The _____ Ocean is famous for the Mariana Trench.

Oceans and Disasters:

Natural ocean events can be devastating:

- **Tsunamis:** Giant ocean waves caused by underwater earthquakes, volcanic eruptions, or landslides.

A Tsunami



A Hurricane

- **Hurricanes:** Big storms from warm waters causing floods and wind damage.

Marine Pollution

- **Marine Pollution:** Plastics, chemicals and waste hurt sea life and ecosystems.

Climate Change and Rising Sea Levels

- **Climate Change:** Earth's warming melts ice and raises sea levels, causing floods and land erosion.

Do you know?

Famous Disasters:

- 2004 Indian Ocean Tsunami: Over 230,000 deaths across 14 countries; Indira Point submerged.
- Hurricane Katrina (2005): Over 1,800 deaths; $125 billion in damage.

Do you know?

Disaster Management in India:

- Disaster Management Act: 23 Dec 2005.
- NIDM, Delhi: Trains officials, police and agencies for disaster preparedness.

Disaster Management

Disaster Management Team of the Government in Action

In-Text Questions

1. ..... is the biggest ocean on the planet.
2. The Earth is termed a water planet since ..... .
3. The Atlantic Ocean broadly resembles the letter ..... in shape.
4. Water is an ..... resource.
5. The Indian Ocean is named after ..... .
6. Oceans hold Earth's ..... % of water found on the surface.
7. ..... is the busiest sea route for trade and commerce.
8. The North Pole is encircled by ..... .
9. NIDM stands for ..... .
10. The Disaster Management Act was announced on ..... .

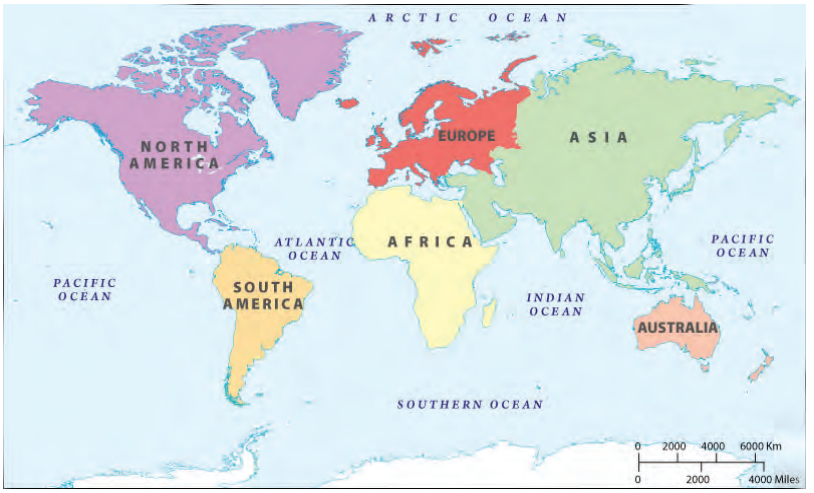
Analyze Thoughts

Explain the importance of the hydrosphere in daily life. Prepare a list of its functions and share them with your friends and teachers. Also explain simple actions you may adopt to conserve water.

CONTINENTS

Continents are large land areas that stand above sea level. They formed over millions of years because of the movement of tectonic plates. Earth's outer layer is made of tectonic plates that are always moving. Long ago, all the continents were joined together as one huge landmass called Pangaea. As the tectonic plates moved, Pangaea slowly broke into smaller parts and formed today's continents. Large oceans separate the continents from each other. Earth has seven continents: Asia, Africa, North America, South America, Antarctica, Europe, and Australia. A world map shows that most of Earth's land is in the Northern Hemisphere. Asia, Europe, and North America is entirely in the Northern Hemisphere, while Australia and Antarctica are completely in

the Southern Hemisphere. For this reason, the Northern Hemisphere is called the Land Hemisphere.



Seven Continents on the Earth

Facts About Continents:

- Asia: Largest continent, home to half the world's population, separated from Europe by the Ural Mountains and Ural River.
- Africa: Second largest continent, called the Dark Continent; lies in all four hemispheres; home to the Nile River and the Sahara Desert.
- North America: Third largest continent, connected to South America by the Isthmus of Panama.
- South America: Fourth largest, almost triangle-shaped; home to the Amazon River.
- Antarctica: Fifth largest, uninhabited, covered with thick ice, called the Frozen Continent.
- Europe: Sixth largest; divided from Asia by the Ural mountains and the Ural river.
- Australia: Smallest continent, also a country, surrounded by water (Island Continent).

Do you know?

India and several other nations have established their research stations in Antarctica. Maitri and Barati are the principal research centres of India located there.

Islands

An island is a piece of land that is fully surrounded by water. Islands are found in oceans, seas, rivers and lakes. They form differently. Some islands are created by eruptions of underwater volcanoes, while others form from the shifting of Earth's tectonic plates. The collection of coral pieces and other natural materials can also help form new islands over time. Islands provide things like fish, wood, and rich soil for farming. They

The Andaman Island

Barren Island: An active volcano

are also popular tourist places due to beautiful views, beaches, and animals. India has more than 1300 islands. Lakshadweep and the Andaman and Nicobar Islands are the main island groups of India. The Lakshadweep Islands are found in the Arabian Sea area. They are known as coral islands because they are formed from corals, which are the hard remains of tiny polyps. They comprise many small coral reefs. The Andaman and Nicobar Islands are located in the Bay of Bengal. Most of these islands are volcanic. Barren Island in the Andaman and Nicobar group is India's only active volcano.

Do you know?

Corals are the hard remains of very small sea creatures known as polyps. When these living polyps die, their hard skeletons remain behind. Gradually, new polyps develop over these skeletons. In this manner, the structure keeps rising layer by layer and finally forms coral islands.

In-Text Questions

1. Biggest continent _____
2. Africa is also known as _____
3. Longest river in the world _____
4. Largest desert on Earth _____
5. Continent covered with ice _____
6. Continent surrounded by water _____
7. India's only active volcano _____

In-Text Questions

Why are continents different from islands?

Continents are surrounded by water as well, but due to their enormous size, they are not called islands. For instance, the smallest continent, Australia, is nearly three times bigger than Greenland, the largest island.

Beautiful landscape of islands

Analyze Thoughts

Consider the different ways humans utilize land.

NEW TERMS

- **Oceans:** Large bodies of saltwater
- **Isthmus:** Narrow land connecting landmasses
- **Hydrosphere:** All water on Earth
- **Atmosphere:** Layer of gases around Earth
- **Continents:** Large land areas on Earth
- **Disaster:** An event causing serious damage
- **Island:** A piece of land surrounded by water
- **Eurasia:** Europe and Asia combined
- **Natural Disasters:** Destructive events caused by nature
- **Earthquake:** Sudden shaking of Earth's surface
- **Flood:** Excess water covering land
- **Disaster Management**: Planning to handle disasters

QUICK TAKE

- Large bodies of salty water that cover most of Earth are known as oceans.
- More than 71% of Earth is covered with water, while only 29% is land, showing that our planet has lots of water.
- Earth has five main oceans: Pacific Ocean, Atlantic Ocean, Indian Ocean, Arctic Ocean and Southern/Antarctic Ocean.
- Since Earth has a large amount of water, it is also called the Blue Planet or Water Planet.
- The Arctic Ocean is the smallest ocean and remains mostly covered with ice during the whole year.
- The Atlantic Ocean looks like the letter S and has many natural harbours for ships and trade.
- The Pacific Ocean is the biggest and deepest ocean on Earth, spreading across many nations.
- In the past, all continents were connected to make one huge landmass called Pangaea.
- Continents are very large pieces of land above sea level where most people live on Earth.
- The constant movement of tectonic plates can cause earthquakes, volcanoes and other natural hazards.
- Lakshadweep and the Andaman and Nicobar Islands are India's main island groups, famous for their beauty and rich biodiversity.
- Earth has seven continents: Asia, Africa, North America, South America, Antarctica, Europe and Australia, each different in size and culture.
- An island is a land area completely surrounded by water, whether it is large or small.
- Asia is the largest continent with the highest population, while Australia is the smallest continent.

EXERCISES

I. OBJECTIVE TYPE QUESTIONS

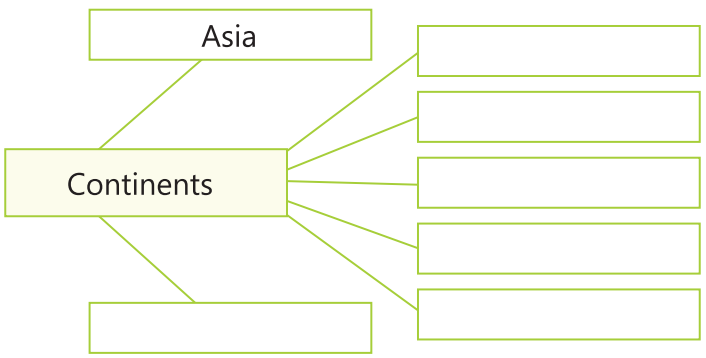
A. Choose the correct answer.

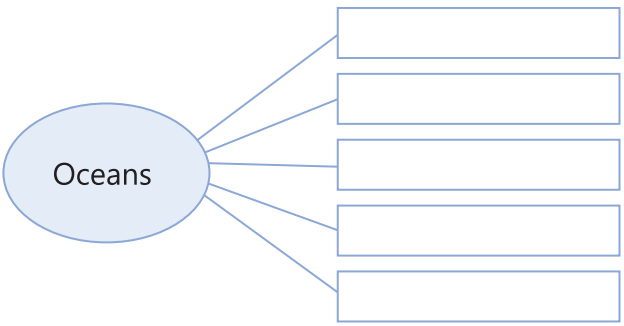
1. Which continent is the biggest continent on Earth?

   (a) Asia ☐ (b) Europe ☐ (c) North America ☐ (d) Australia ☐

2. Which place was covered by seawater during the 2004 Tsunami?
   (a) Nanda Devi ☐ (b) Palk Strait ☐ (c) Indira Point ☐ (d) Port Blair ☐
3. The Government of India approved the Disaster Management Act in which year?
   (a) 2004 ☐ (b) 2003 ☐ (c) 2006 ☐ (d) 2005 ☐
4. Which mountain range acts as a natural boundary between Asia and Europe?
   (a) Himalayas ☐ (b) Ural ☐ (c) Atlas ☐ (d)All of these ☐
5. A disaster is a ..... event that causes major damage or loss of life.
   (a) natural ☐ (b) human-made ☐
   (c) both a and b ☐ (d) none of these ☐

B. Fill in the blanks.





C. Choose the option that correctly matches Column A with Column B.

Column A	Column B
1. Australia	(i) Frozen continent
2. Antarctica	(ii) Dark continent
3. Asia	(iii) Roughly triangular in shape
4. Africa	(iv) Smallest continent
5. South America	(v) Largest continent

(a) 1.-(i), 2.-(iii), 3.-(iv), 4.-(ii), 5.-(v)
(b) 1.-(i), 2.-(iv), 3.-(ii), 4.-(v), 5.-(iii)
(c) 1.-(iv), 2.-(i), 3.-(ii), 4.-(v), 5.-(iii)
(d) 1.-(iv), 2.-(i), 3.-(v), 4.-(ii), 5.-(iii)

D. Write 'T' for True or 'F' False.

1. The combined landmass of Asia and Europe is known as Eurasia. ☐
2. The Indian Ocean is named after the country of India. ☐
3. The River Nile flows through Australia. ☐
4. The Andaman and Nicobar Islands are located in the Arabian Sea. ☐
5. The Mariana Trench is the deepest point in the Atlantic Ocean. ☐

E. Assertion-Reason

For questions 1 to 4, two statements are provided—one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer for each question from the options (a), (b), (c), and (d) as mentioned below:

1. **Assertion (A)** : The Atlantic Ocean has an 'S' shape.
   Reason (R) : The Arctic Ocean is the largest of all oceans.
2. **Assertion (A)** : Earth is referred to as the "Blue Planet."
   Reason (R) : Water covers more than 70% of the Earth's surface.
3. **Assertion (A)** : Europe is known as the "Dark Continent."
   Reason (R) : Australia is the smallest continent in the world.
4. **Assertion (A)** : The Lakshadweep Islands are found in the Arabian Sea.
   Reason (R) : These islands are called coral islands because they are formed from coral skeletons.

(a) Both Assertion (A) and Reason (R) are correct and Reason explains Assertion.

(b) Both Assertion (A) and Reason (R) are correct but Reason doesn't explain Assertion.

(c) Assertion (A) is correct but Reason (R) is incorrect.

(d) Assertion (A) is incorrect but Reason (R) is correct.

II. SHORT ANSWER TYPE QUESTIONS

1. What are the causes of disasters occurring in the oceans?
2. What makes Earth known as the Blue Planet?
3. List three characteristics of the continent of Asia.
4. Write a brief description of marine life in the oceans.
5. Explain why the ocean's surface level remains constant across the globe.

III. LONG ANSWER TYPE QUESTIONS

1. Discuss two major oceanic disasters.
2. Explain why the Atlantic Ocean serves as the world's main trade and commerce route.
3. Write about the significance of oceans in our ecosystem.
4. Describe the plant and animal life found in the oceans.
5. Explain the process of island formation.

IV. PICTURE STUDY

1. What natural event is shown in the picture?
2. How can people stay safe during such a natural event?
3. Why are coastal areas more affected by these events?

V. Read the given paragraph and answer the following questions.

Earth is known as the Blue Planet because more than 71% of its surface has water, mainly salty oceans (97.3%), leaving only 0.03% as freshwater for people. Oceans are connected, control climate, support sea life and join continents for trade and travel. Continents are huge land areas made by tectonic changes; long ago, all land was together as Pangaea, then it slowly split into seven continents. Islands, made by volcanoes or coral growth, give resources, fertile land and tourism opportunities. This shows that freshwater conservation and ocean protecton are very important.

1. If only 0.03% of Earth's water is available as freshwater for people, what does this show about our duty?
   - (a) Use freely ☐ (b) Save water ☐
   - (c) Clean oceans ☐ (d) Endless rivers ☐
2. Oceans are connected and keep the same sea level. Why is this useful for human activities?
   - (a) Measure land ☐ (b) Trade and travel ☐
   - (c) More freshwater ☐ (d) Larger islands ☐
3. The splitting of Pangaea into seven continents teaches us that:
   - (a) Seven continents ☐ (b) Earth changes ☐
   - (c) Oceans form ☐ (d) Coral islands ☐
4. Islands are created by volcanic action or coral growth. What benefit do they give humans?
   - (a) Decrease continents ☐ (b) Resources, tourism ☐
   - (c) More freshwater ☐ (d) Stop tectonic changes ☐

Continents and Oceans Fact File

1 Each student prepares a booklet with key facts:
- Continent info: area, population, key countries, landmarks and unique features.
- Size, depth and importance of each ocean.

2 Present as a classroom exhibition.

MAP SKILL

- Use the words from the box below to label the continents and oceans of the world.
  (Hint: Continents are the yellow boxes and oceans are the pink boxes on the globe.)

Asia	North America	Africa
Arctic Ocean	Atlantic Ocean	Indian Ocean
South America	Europe	Antarctica
Australia	Pacific Ocean	Southern Ocean